WORKED SOLUTION

STEP II 1997, Question 14

Probability and Statistics • Poisson Processes

STEP II 1997, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q14

The key idea

Despite the mechanics setting, this is a Poisson process question. The kinematics just tells us which vehicles count: a faster vehicle catches me inside the tunnel only if it enters less than a certain time after me. Once we know that cut-off, everything becomes a question about Poisson counts. In the second half we use the fact that a Poisson stream whose vehicles are labelled fast or slow at random splits into two independent Poisson streams. We also need to notice that one slow vehicle shields me from every fast vehicle behind it.

Solution

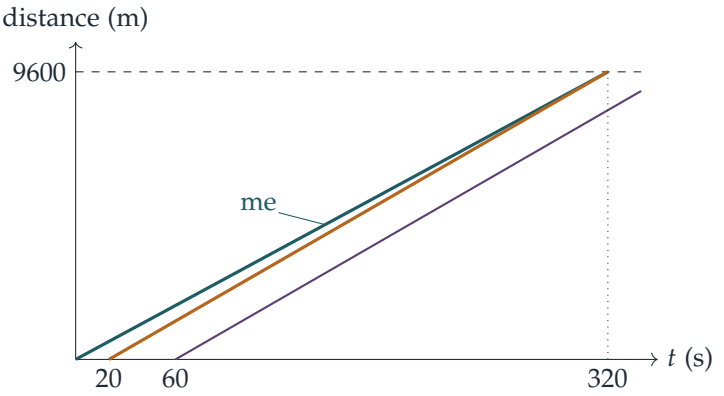
Throughout we model vehicles as particles (their lengths are ignored), so a vehicle that catches up simply travels at the speed of the one in front from then on. Measure time t in seconds from the moment I enter. Vehicles enter at a rate of 6 per minute, which is 0.1 per second, so the number entering in any interval of length τ seconds is $\mathrm{Po}(0.1\tau)$, and the counts in disjoint intervals are independent.

The first part: who can catch me?

I take $9600/30 = 320\,\mathrm{s}$ to get through the tunnel. Consider a vehicle at $32\,\mathrm{m\,s^{-1}}$ entering at time $t > 0$. Until it catches me, my distance in is $30s$ at time s and its distance is $32(s-t)$. These are equal when $2s = 32t$, i.e. $s = 16t$, at a distance of $480t$ metres into the tunnel. So it catches me before I leave exactly when

$$480t \leq 9600 \iff t \leq 20.$$

(Equivalently: it needs $9600/32 = 300\,\mathrm{s}$ to get through, so it reaches the exit no later than me exactly when $t + 300 \leq 320$.)



The distance–time graph shows me (teal), a fast vehicle entering at $t = 20$ (amber) and one entering at $t = 60$ (plum). The amber vehicle meets me just as we both reach the exit; any fast vehicle entering later, like the plum one, is still behind me when I leave.

Every other vehicle travels at $32\,\mathrm{m\,s^{-1}}$, so nothing can hold up a vehicle entering in $(0, 20]$ before it reaches me: any vehicle ahead of it, apart from me, is also travelling at $32\,\mathrm{m\,s^{-1}}$ until it too reaches me. Vehicles that entered before me are ahead of me and irrelevant. So the queue behind me when I leave consists of exactly the vehicles entering in $(0, 20]$, whose number is $\mathrm{Po}(0.1 \times 20) = \mathrm{Po}(2)$. The expected number of vehicles in the queue behind me is

$$\boxed{2}.$$

The second part: exactly two enter and catch me

Now each other vehicle is fast ($32\,\mathrm{m\,s^{-1}}$) or slow ($30\,\mathrm{m\,s^{-1}}$) with probability $\frac{1}{2}$, independently of everything else.

Ask yourself Does every fast vehicle entering within 20 seconds still catch me?

No. A slow vehicle entering at time $t' > 0$ keeps a constant gap of $30t'$ metres behind me, so it never catches me. A fast vehicle entering after it catches the slow vehicle (or is blocked by something behind it) and is stuck at $30\,\mathrm{m\,s^{-1}}$ from then on, so it never reaches me either. A vehicle entering in $(0, 20]$ therefore catches me if and only if it is fast *and* every vehicle entering between me and it is fast. A fast vehicle that enters after $t = 20$ cannot catch me even if nothing is in its way.

Let $K \sim \mathrm{Po}(2)$ be the number of vehicles entering in $(0, 20]$. For the event in this part, exactly two vehicles enter in that window and both catch me. That needs $K = 2$ with both vehicles fast:

$$\mathrm{P} = \mathrm{P}(K = 2) \times \left(\tfrac{1}{2}\right)^2 = \frac{2^2 \mathrm{e}^{-2}}{2!} \cdot \frac{1}{4} = \boxed{\tfrac{1}{2}\mathrm{e}^{-2}} \approx 0.0677.$$

The final part: exactly two queuing behind me

Now any number of vehicles may enter in the first 20 seconds. List the K vehicles entering in $(0, 20]$ in order of entry. By the argument above, the queue behind me at the exit is the unbroken run of fast vehicles at the start of this list. (No vehicle entering after $t = 20$ can join, since it can never catch me.) So there are exactly two vehicles in the queue if and only if

- $K = 2$ and both are fast: probability $\frac{1}{4}$ given $K = 2$; or
- $K \geq 3$ and the first two are fast and the third is slow: probability $\frac{1}{8}$ given $K = k \geq 3$.

Hence

$$\begin{aligned} \mathrm{P}(\text{queue of two}) &= \tfrac{1}{4}\,\mathrm{P}(K = 2) + \tfrac{1}{8}\,\mathrm{P}(K \geq 3) \\ &= \tfrac{1}{4} \cdot 2\mathrm{e}^{-2} + \tfrac{1}{8}\left(1 - \mathrm{e}^{-2} - 2\mathrm{e}^{-2} - 2\mathrm{e}^{-2}\right) \\ &= \tfrac{1}{2}\mathrm{e}^{-2} + \tfrac{1}{8} - \tfrac{5}{8}\mathrm{e}^{-2} \\ &= \boxed{\tfrac{1}{8}\left(1 - \mathrm{e}^{-2}\right)} \approx 0.108. \end{aligned}$$

This is larger than the previous answer, as it must be: that event is one of the ways of getting a queue of two.

> A neat alternative: split the stream into fast vehicles and slow vehicles, two independent Poisson processes each of rate $0.05\,\mathrm{s}^{-1}$. A queue of exactly two means that, in the first 20 seconds, the first two arrivals are fast and the next arrival (if there is one) is slow. Thinking in terms of these two streams is often the quickest route in Poisson questions with labelled events.

Checks

- Limiting case: if every vehicle were slow, nobody could catch me, and indeed the queue formula gives 0 when the probability of being fast is 0.
- With probability p of being fast in place of $\frac{1}{2}$, the same argument gives $p^2\mathrm{P}(K = 2) + p^2(1 - p)\mathrm{P}(K \geq 3)$. At $p = 1$ this is $\mathrm{P}(K = 2) = 2\mathrm{e}^{-2}$, matching the first part, where the queue length is exactly $\mathrm{Po}(2)$.
- We simulated 4×10^5 runs of the traffic with no overtaking. They gave a mean queue of 2.003 in the first part, and probabilities 0.0674 and 0.1076 in the second, against the exact 0.0677 and 0.1081.

Common slips

- Forgetting that a slow vehicle shields me from every fast vehicle behind it, and answering $\mathrm{P}(\text{exactly two fast in 20 s}) = \frac{1}{2}\mathrm{e}^{-1}$ for the final part.

- Using the rate 6 per minute with a time in seconds: in 20 seconds the mean is 2, not 120.
- Assuming the queue grows from vehicles entering at any time: only those entering within the first 20 seconds can reach me.

Beyond STEP

The second part uses *thinning*: if each event of a Poisson process of rate λ is kept independently with probability p, the kept and discarded events form independent Poisson processes of rates $p\lambda$ and $(1-p)\lambda$. Queues of vehicles that cannot overtake are a classic model in traffic flow theory, where slow vehicles produce "platoons" of the kind we found here. For Poisson processes and thinning, see Grimmett and Welsh, *Probability: An Introduction*.

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